

class 12

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section 1. proportion of G/G in a population

downloaded a CSV file from Ensembl https://useast.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:43440445-43443746;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel

here we read this CSV file

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mx1)
```

	Sample..Male.Female.Unknown.	Genotype..forward.strand.	Population.s.	Father
1	NA19648 (F)	A A	ALL, AMR, MXL	-
2	NA19649 (M)	G G	ALL, AMR, MXL	-
3	NA19651 (F)	A A	ALL, AMR, MXL	-
4	NA19652 (M)	G G	ALL, AMR, MXL	-
5	NA19654 (F)	G G	ALL, AMR, MXL	-
6	NA19655 (M)	A G	ALL, AMR, MXL	-
Mother				
1	-			
2	-			
3	-			
4	-			
5	-			
6	-			

```
table(mx1$Genotype..forward.strand.)
```

A A	A G	G A	G G
22	21	12	9

```
table(mx1$Genotype..forward.strand.)/nrow(mx1)
```

```
      A|A      A|G      G|A      G|G
0.343750 0.328125 0.187500 0.140625
```

now let's look at a different population. I picked the GBR.

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

find proportion of G|G

```
round(table(gbr$Genotype..forward.strand.)/nrow(gbr)*100,2)
```

```
      A|A      A|G      G|A      G|G
25.27 18.68 26.37 29.67
```

This variant that is associated with childhood asthma is more frequent in the GBR population than the MKL population.

Lets now dig into this further.

section 4: population scale analysis

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

how many samples do we have?

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head (expr)
```

```
      sample geno      exp
1 HG00367   A/G 28.96038
2 NA20768   A/G 20.24449
3 HG00361   A/A 31.32628
4 HG00135   A/A 34.11169
5 NA18870   G/G 18.25141
6 NA11993   A/A 32.89721
```

```
nrow(expr)
```

```
[1] 462
```

. Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
table(expr$geno)
```

```
A/A A/G G/G  
108 233 121
```

```
library(ggplot2)
```

Q14:

let's make a boxplot

```
ggplot(expr) + aes(geno, exp, fill=geno)+  
  geom_boxplot(notch=TRUE)
```

